Women among students and teachers in Indian higher education

National figures from the All India Survey on Higher Education (AISHE), 2019-20 to 2023-24

1. Summary

- **Women are now about half of all students.** In 2023-24 women were 49.7% of the 45.0 million students enrolled, up from 49.0% in 2019-20 (it dipped to 47.8% in 2021-22). They are 49.5% of undergraduates, 56.2% of postgraduates and 49.8% of PhD students, up from 45.0% in 2019-20.
- **Women are a smaller share of teachers: 44.9%** of 1.73 million in 2023-24 (up from 42.5%). The share falls with seniority: 45.4% of assistant professors, 38.8% of associate professors and **31.3% of professors**.
- **The gap between students and professors is about 18 percentage points** (50% of PhD students against 31% of professors). Women's share of professors rose by 3.8 points in four years, which is faster than the other ranks, but from a low base.
- **The gap is widest in the institutions most like IISER.** Institutes of National Importance (which include the IISERs, IITs and NITs) had 22.1% women teachers in 2023-24, and central universities 27.8%, against 41 to 54% in state universities and affiliated colleges.
- **Science and engineering differ.** Women are 53.4% of undergraduate and 61.4% of postgraduate science students, but 31.1% of undergraduate engineering students. In engineering the share of women among PhD students (37.0%) is higher than at undergraduate level and has risen from 31.7% in 2019-20.
- **States differ more than levels do.** Women are 27% of teachers in Bihar and 32% in Uttar Pradesh, against 64% in Kerala and 62% in Punjab, even though women are 49 to 58% of students in all of these. West Bengal, our own state, has 37.4% women teachers, fifth lowest of the larger states.

2. Source and method

All numbers come from the AISHE final reports for 2019-20, 2021-22, 2022-23 and 2023-24, published by the Ministry of Education, using the tables below. The 2020-21 report was not available to us, so it is missing from every trend, and the first step in each is two years (2019-20 to 2021-22). Changes are treated as changes over the stated period, not annual rates.

Quantity	AISHE table	What it counts
Students by level and gender	Table 6	Enrolment, *estimated for all institutions, including those that did not respond
Disciplines (UG; PhD, M.Phil., PG)	Tables 12 and 13	Enrolment by broad subject area
Gross Enrolment Ratio and Gender Parity Index	Tables 19 and 20	Enrolment as % of the 18 to 23 age group, using population projections based on the 2011 census
Teachers by rank and gender	Table 22	Teachers in institutions that *responded; not scaled up
Teachers by type of institution	Table 29	Teachers in universities and in their affiliated colleges

"Women's share" is women divided by women plus men (transgender counts are not in the tables we used). Because student counts are estimates and teacher counts are from responding institutions, we compare **shares**, not absolute numbers of students and teachers, and we do not calculate student-to-teacher ratios.

3. Women as students

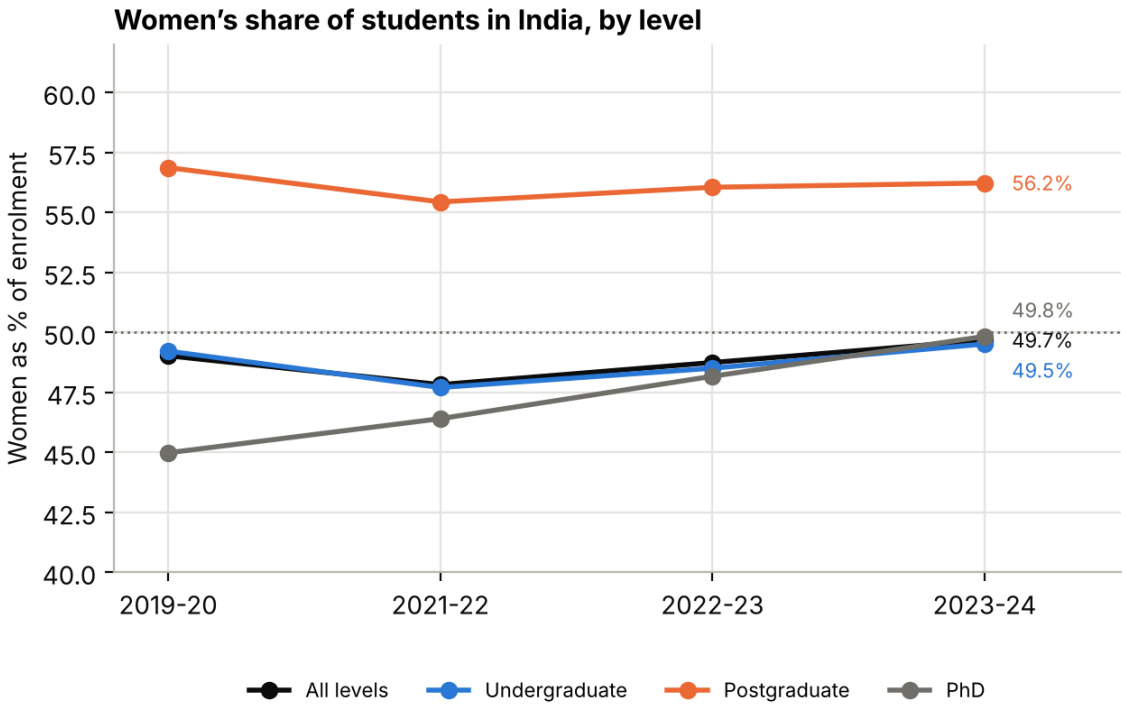


Figure 1. Women as a percentage of enrolment, by level. Women outnumber men at postgraduate level in every year and are close to parity in undergraduate and PhD enrolment.

Level	2019-20	2021-22	2022-23	2023-24
Undergraduate	49.2%	47.7%	48.5%	49.5%
Postgraduate	56.9%	55.4%	56.1%	56.2%
M.Phil.	62.2%	64.4%	60.7%	68.0%

Level	2019-20	2021-22	2022-23	2023-24
PhD	45.0%	46.4%	48.2%	49.8%
Diploma	34.9%	36.2%	38.4%	40.0%
All levels	49.0%	47.8%	48.7%	49.7%

Women's share of **PhD enrolment** rose steadily, from 45.0% to 49.8%. The postgraduate share has stayed between 55% and 57%. M.Phil. is a shrinking programme (the share is high, but numbers are small) and could not be used as a trend. Diploma courses, which are mostly technical, are the one level where women are well below half (40.0%).

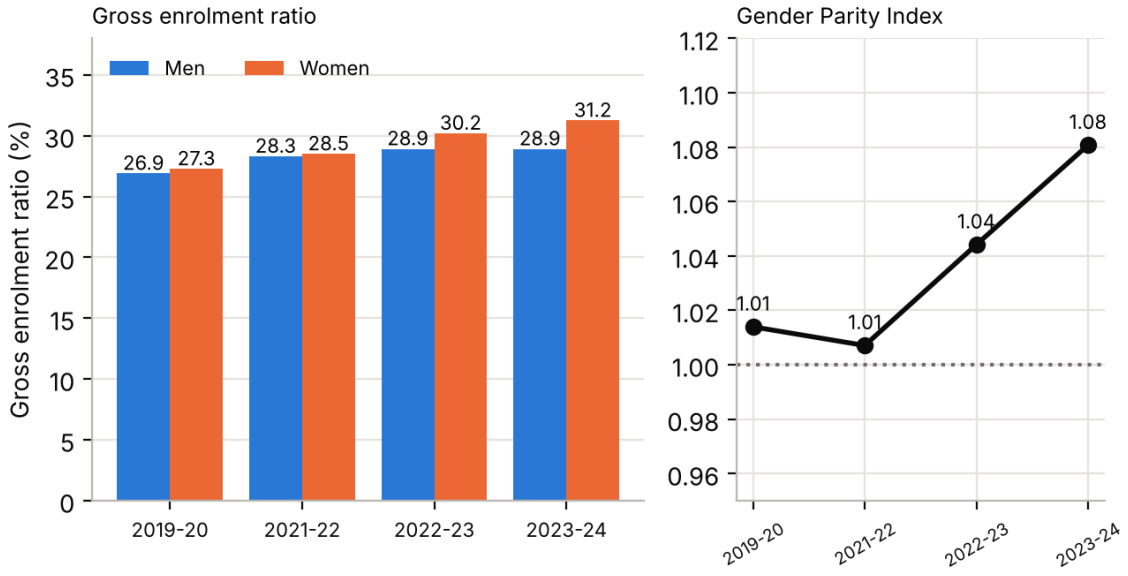


Figure 2. Gross Enrolment Ratio (left) and Gender Parity Index (right). GER is enrolment as a percentage of the 18 to 23 age group; GPI is the women's GER divided by the men's.

The women's GER rose from 27.3 to 31.2, and the men's from 26.9 to 28.9. The GPI, which was 1.01 in 2019-20, reached 1.08 in 2023-24: among people of college age, more women than men are enrolled. Both ratios are based on population projections from 2011, so the absolute level is uncertain, and the trend is more reliable than any single value.

4. Women as teachers

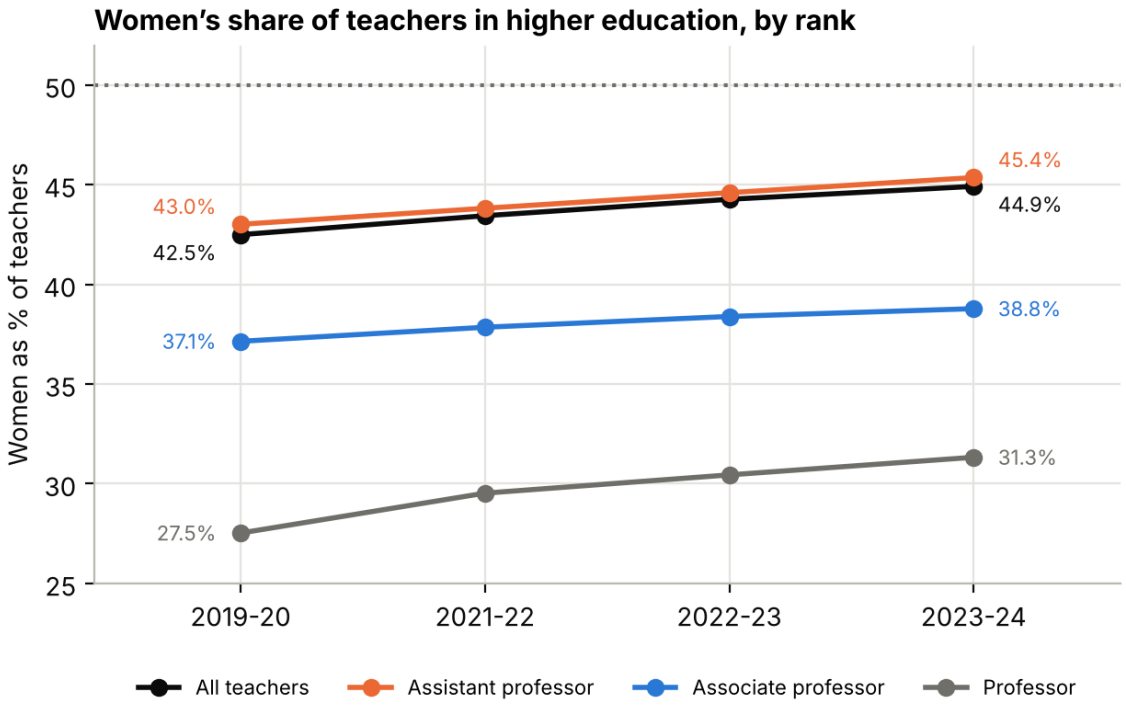


Figure 3. Women as a percentage of teachers, by rank. The dotted line marks parity.

Rank	2019-20	2021-22	2022-23	2023-24
Professor	27.5%	29.5%	30.4%	31.3%
Associate professor	37.1%	37.9%	38.4%	38.8%
Assistant professor	43.0%	43.8%	44.6%	45.4%
All teachers	42.5%	43.4%	44.3%	44.9%

In 2023-24, women held 52,490 of 167,559 professor posts, 65,674 of 169,332 associate professor posts and 528,197 of 1,164,360 assistant professor posts. Women are also 67% of demonstrators and tutors, the lowest-ranking and least secure teaching position in the table, and the only category where they are a clear majority.

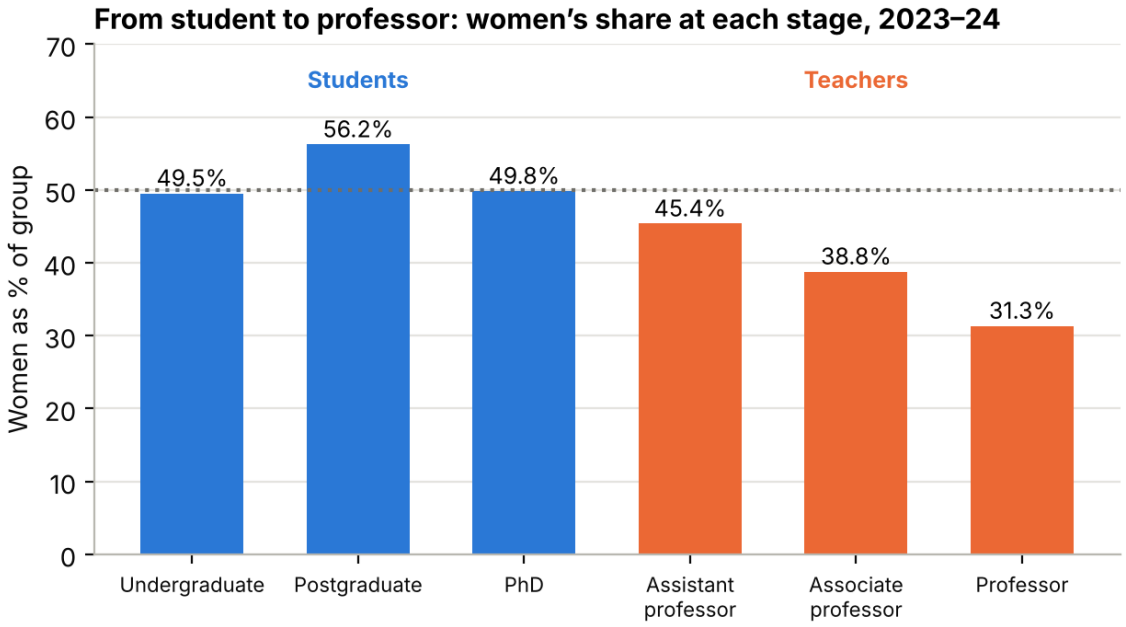


Figure 4. The pipeline in 2023-24: women's share at each stage from undergraduate student to professor. The two groups are not the same cohort; this is a cross-section of one year.

Our interpretation of the pipeline:

1. The figure is a snapshot of one year, not a follow-up of a cohort
2. Women's share of professors reflects who was hired 20 to 30 years ago, when women were a much smaller share of PhD students
3. So the 31% will rise in future even without further change, but only slowly, and the rise in recent years (27.5% to 31.3%) cannot say whether women are being promoted at the same rate as men today.

5. By type of institution

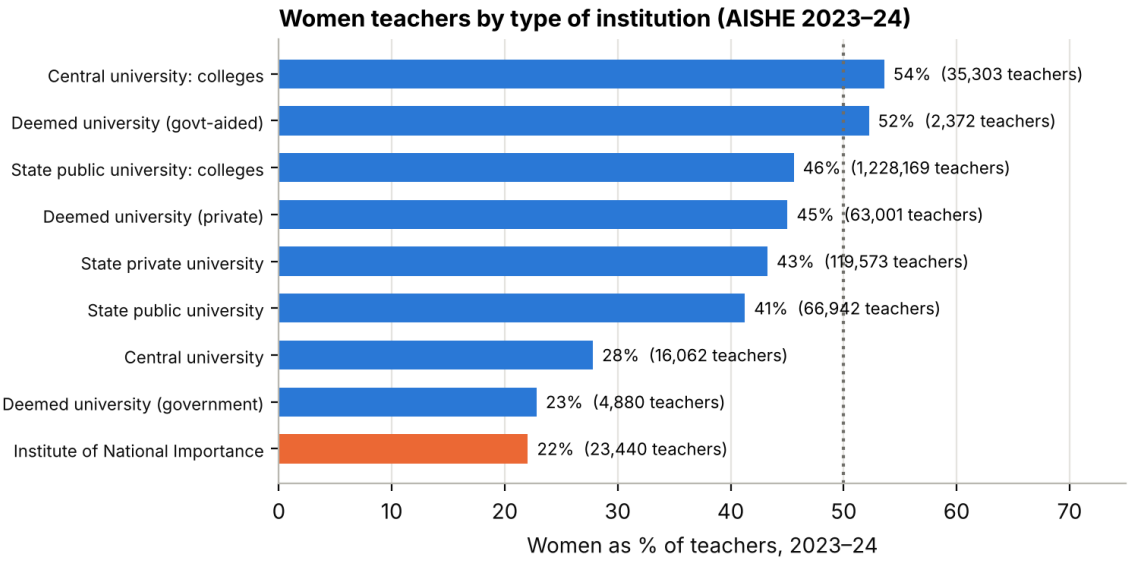


Figure 5. Women as a percentage of teachers by type of institution, 2023-24.

Women are 22.1% of teachers in Institutes of National Importance. This is the group that includes IISERs, IITs, NITs and AIIMS, and it is the closest national comparison for IISER Kolkata. The share in these institutes rose from 20.3% (2019-20) to 22.1% (2023-24), and is lower at higher ranks: 24.4% of assistant professors, 20.6% of associate professors and 17.5% of professors. Government deemed universities (22.8%) and central universities (27.8%) are close behind. The highest shares, in private universities and affiliated colleges, are in institutions that rely more on teaching than research, and often on lower pay and weaker job security.

This group also includes institutes with large engineering and medical faculties, so it is a rough comparison for IISER Kolkata, not a benchmark for what is possible. The NIRF summary in our earlier note gives the IISER Kolkata figure.

6. By discipline

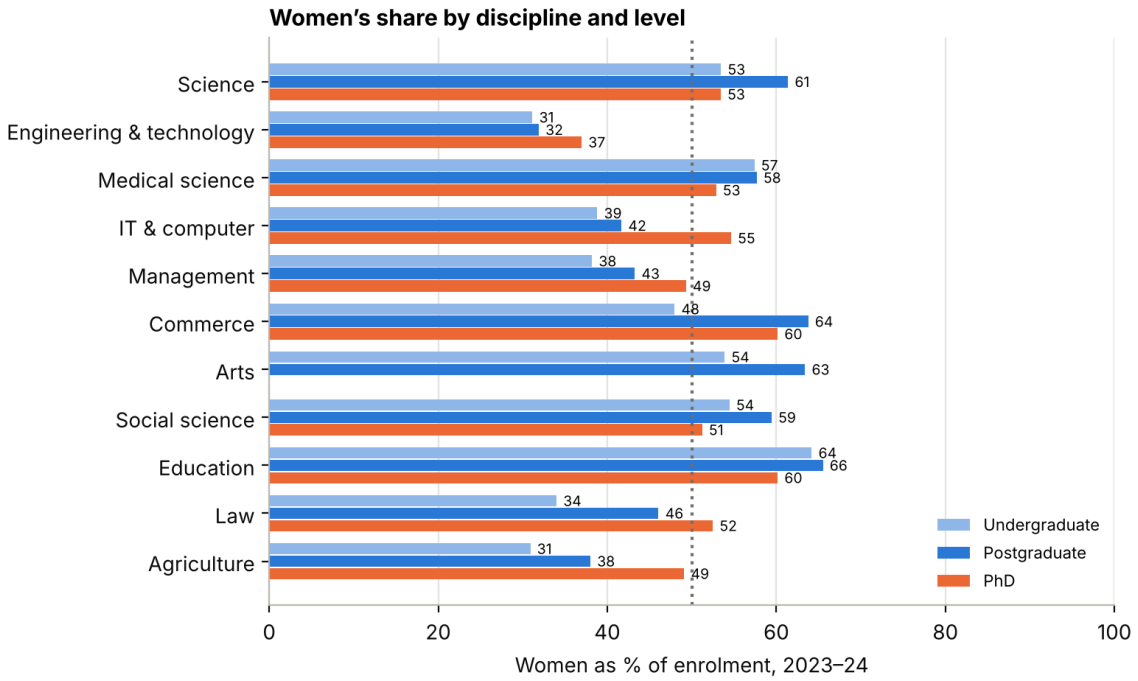


Figure 6. Women as a percentage of enrolment by broad discipline and level, 2023-24. The dotted line is parity.

Women's share varies much more by discipline than by level. They are a majority in education, medical science, arts, social science and (at postgraduate level) commerce, and a minority in engineering, agriculture, IT and computer, law (at undergraduate level) and management. In most disciplines, women's share is higher at postgraduate level than at undergraduate level. In science and medical science it falls from postgraduate to PhD level (science: 61.4% to 53.4%).

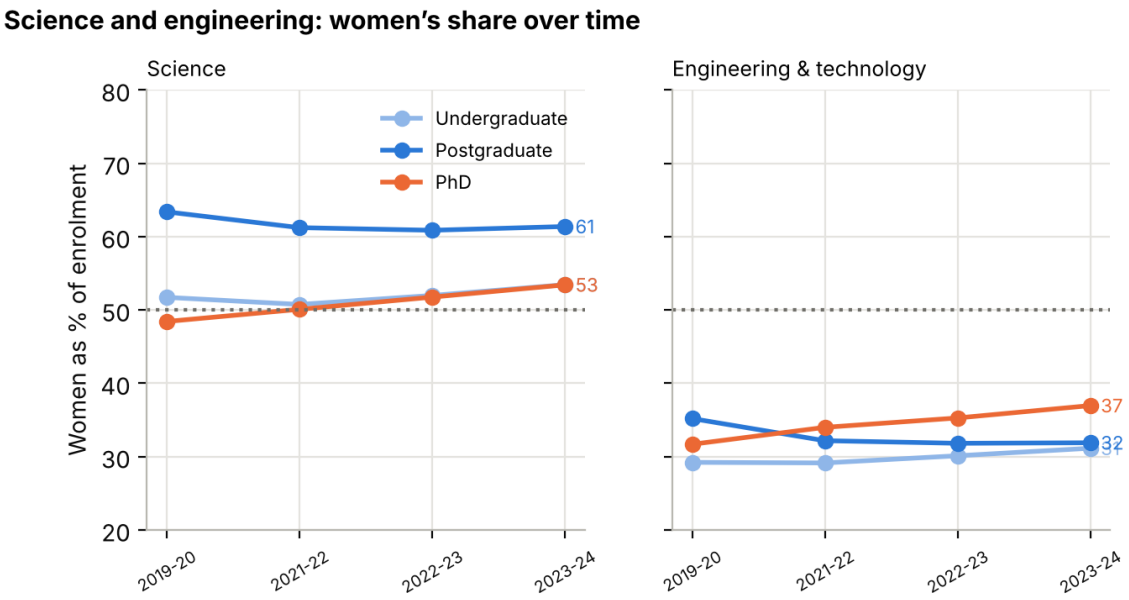


Figure 7. Women's share of undergraduate, postgraduate and PhD enrolment in science and in engineering and technology, 2019-20 to 2023-24.

Science (which includes the basic sciences taught at IISER) had 53.4% women at undergraduate, 61.4% at postgraduate and 53.4% at PhD level in 2023-24. The PhD share rose from 48.4% in 2019-20. **Engineering and technology** is where women are least represented, at 31.1% of undergraduates, though the PhD share rose from 31.7% to 37.0%. Note that the PhD numbers in engineering are much smaller than the undergraduate numbers, so the PhD share is more sensitive to who enrols.

7. By state and union territory

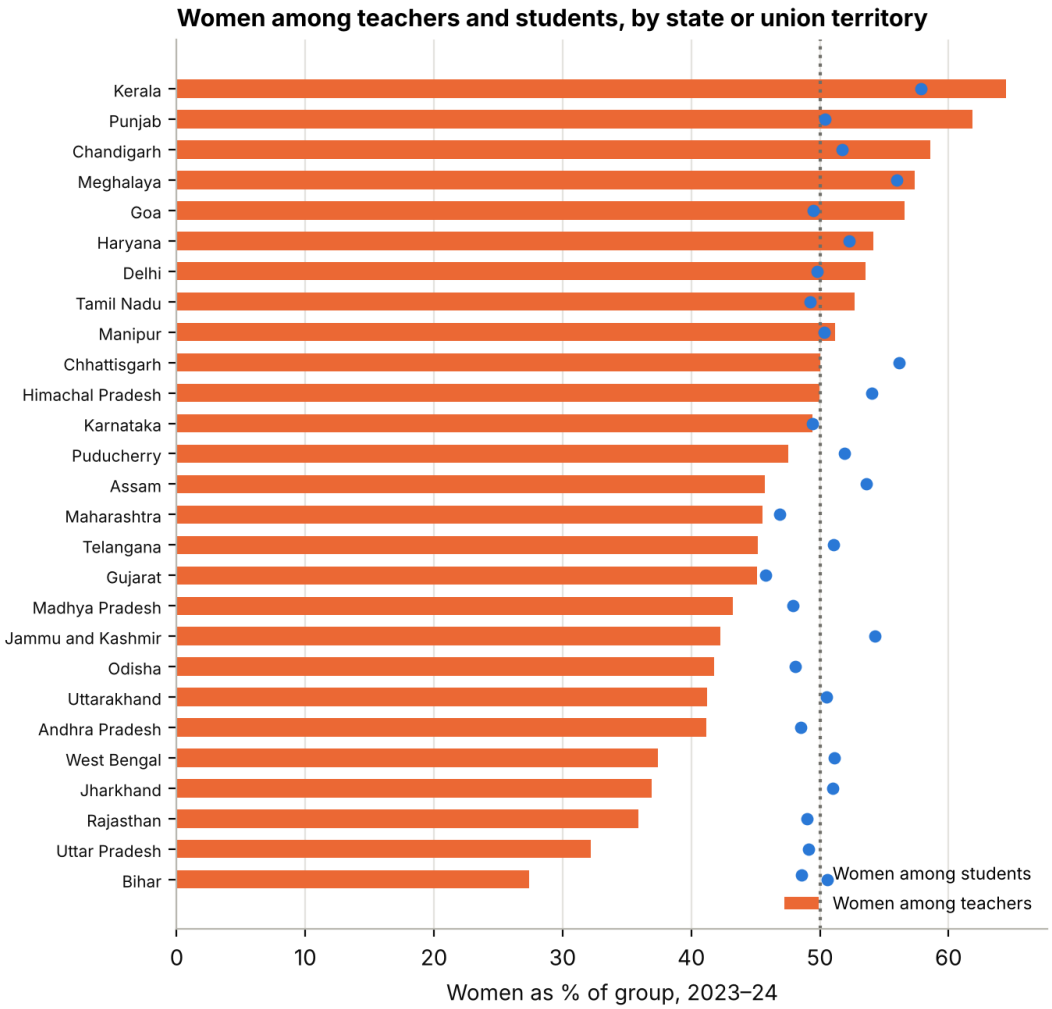


Figure 8. Women among teachers (bars) and among students (dots) in 2023-24, for the 27 states and union territories with at least 3,000 teachers. The dotted line is 50%.

Women's share of students is between 46% and 58% in all of these 27 states and union territories, but the share of teachers ranges from 27% to 64%. In 17 of 27, the share of women among teachers is lower than their share among students.

- **Highest teacher shares:** Kerala (64.5%), Punjab (61.9%), Chandigarh (58.6%), Meghalaya (57.4%), Goa (56.6%).
- **Lowest teacher shares:** Bihar (27.4%), Uttar Pradesh (32.2%), Rajasthan (35.9%), Jharkhand (36.9%), West Bengal (37.4%).

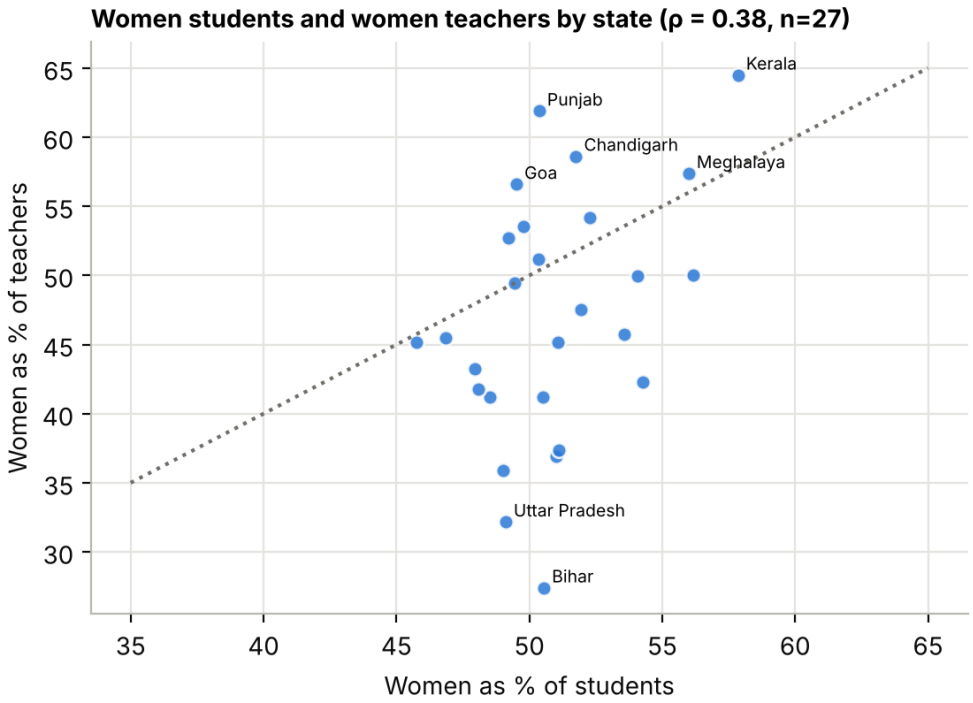


Figure 9. Each dot is a state or union territory. The dotted diagonal is where the two shares are equal; states below it have a lower share of women among teachers than among students.

States with more women students tend to have more women teachers (Spearman $\rho = 0.38$, $n = 27$, $p = 0.05$), but the link is weak and could be due to chance. Kerala is high on both; Bihar has a student share like the national average and the lowest teacher share. The state-level pattern therefore does not look like a simple result of how many women study.

A state's teacher count depends on how many of its institutions responded to the survey and how teachers are counted, so small differences between states should not be over-read. The very large gaps (Bihar and Uttar Pradesh against Kerala and Punjab) are much bigger than that.

8. Limits

- **2020-21 is missing**, so trends skip a year and the pandemic years are poorly represented.
- **Estimated and reported figures are mixed.** Enrolment (Table 6) is an estimate for all institutions; teacher counts (Table 22) are only for institutions that responded. Shares are comparable; totals are not.
- **Binary gender categories.** AISHE reports male, female and (in some tables) transgender, and we used only male and female.
- **Snapshots, not cohorts.** None of the figures follow individuals over time, so they do not show drop-out or promotion rates.
- **Disciplines are broad.** "Science" includes subjects from physics to nutrition, so it is not the same as the mix at IISER.
- **Population projections.** GER and GPI use 2011 census projections, which are not exact.
- **No caste or disability breakdown.** AISHE reports these separately; this report is limited to gender.

Appendix: all states and union territories, 2023-24

State / UT	Women teachers	All teachers	Women % (teachers)	Women % (students)
Kerala	43,676	67,723	64.5	57.9
Punjab	32,082	51,839	61.9	50.4
Nagaland	1,690	2,872	58.8	55.0
Chandigarh	2,116	3,610	58.6	51.8
Meghalaya	2,417	4,213	57.4	56.0
Goa	2,397	4,236	56.6	49.5
Haryana	24,336	44,928	54.2	52.3
A & N Islands	75	139	54.0	50.8
Delhi	13,057	24,377	53.6	49.8
Tamil Nadu	115,206	218,656	52.7	49.2
Manipur	3,505	6,845	51.2	50.3
Chhattisgarh	13,013	25,999	50.0	56.2
Himachal Pradesh	5,668	11,347	50.0	54.1
Mizoram	1,134	2,291	49.5	50.4
Karnataka	77,224	156,162	49.5	49.4
Lakshadweep	33	67	49.2	66.4
D & N Haveli and Daman & Diu	317	653	48.5	49.2
Puducherry	3,921	8,246	47.5	51.9
Assam	12,847	28,083	45.8	53.6
Arunachal Pradesh	1,088	2,390	45.5	48.1
Maharashtra	84,868	186,465	45.5	46.9
Telangana	38,208	84,568	45.2	51.1
Gujarat	31,261	69,247	45.1	45.8
Sikkim	735	1,693	43.4	50.9
Madhya Pradesh	38,861	89,840	43.3	47.9
Jammu and Kashmir	5,078	12,016	42.3	54.3
Odisha	20,607	49,361	41.8	48.1
Uttarakhand	8,271	20,064	41.2	50.5
Andhra Pradesh	44,612	108,354	41.2	48.5
Ladakh	139	352	39.5	63.7
West Bengal	29,243	78,203	37.4	51.1
Tripura	1,117	2,989	37.4	50.1
Jharkhand	6,712	18,180	36.9	51.0
Rajasthan	33,010	91,920	35.9	49.0
Uttar Pradesh	67,292	209,212	32.2	49.1
Bihar	12,371	45,154	27.4	50.5

Student figures for states are AISHE estimates. States and union territories with fewer than 3,000 teachers (not shown in Figures 8 and 9) have small numbers and large year-to-year changes.